Table 1: Part of the codebook categorizing investigative methods with binary coding (reproduced from Tippe and Tippe, 2024).

Code	Description and examples*	Example quotes
Physical	Surveillance of physical activities * Package interception * Camera recordings * Surveillance by agents * Surveillance by confidential informants * Audiorecordings from hidden devices * Community supervision (parole, supervised release, probation) * Meetup with disguised agents * Suspicious package	*A selection of photographs taken during the surveillance are included below:* *Law enforcement subsequently observed BURGAMY walking with what appeared to be multiple United States Postal Service mailing envelopes* *The Postal Inspector then conducted physical surveillance of the Post Office in Cumming, Georgia, on or about August 6, 2020.*
Online	Surveillance of online activities * Live monitoring of accounts * Wiretapping calls * Identifying BitTorrent clients * Pen trap * Network monitoring * GPS tracking * Phone ping	*a court in the Netherlands authorized a wiretap of the computer assigned to the IP address ending in 247, which began on or about May 24, 2016. United States authorities received a copy of the wiretap data…* *a pen register order for the Buchta Comcast Account was issued in the Northern District of Illinois, and thereafter renewed three times. The pen register collection reflected extensive communication between the Buchta Comcast Account and IP addresses ending in …* *Pursuant to a Pen Register/Trap and Trace authorized by the Honorable Judge John F. Anderson for PAGAN's home internet on December 18, 2020, records show network connections from PAGAN's home internet to TOR nodes on multiple days which indicates the use of TOR which is needed to connect to the darkweb.*
Pseudonyms	Linking pseudonyms across platforms * Usernames, mail addresses and other identifiers	*During the course of this investigation, investigators reviewed darknet marketplaces and observed vendor accounts with the moniker addy4cheap offering drugs for sale on the Empire Market and Cryp0nia* *Addy4cheap offered Adderall for sale on the Empire Market and Cryp0nia for quantities ranging from one to over 1000*
Cryptographic keys	Linking cryptographic keys across platforms * DSA login keys * Crypto (hardware) wallets * GPG/PGP keys * Certificates * SSH keys	*The PGP public keys listed for H00k3d on all of the aforementioned sites were identical* *It was discovered that the "SafeServe" accounts on all of these markets used the same PGP key, indicating that the same individual or individuals operated each of the accounts.*
Private information	Information normal persons cannot access * Drivers license register * Jail records * Vehicle registration * Information from confidential informants * Information from other investigations (seized servers, seized devices) * Postal records * Tax databases * Biometric databases * Employment records * Law enforcement databases * Company records without account	*A search of the Colorado Department of Motor Vehicle revealed Gregory Lopez, DOB-XX-XX-1993 with a reported home address in Colorado Spring, Colorado.* *The photo of the male provided by HSI Newark appeared to be the same individual in the driver's license photo of the defendant.* *A review of U.S. Postal Service ("USPS") business records indicated that the aforementioned indivual had received three packages* *Based upon reviews of law enforcement databases*
Public information	Information normal persons can access * Online platforms accounts (i.e. StackOverflow, Instagram, Reddit) * Open source databases * Company registers (which companies exists, who owns them etc.) * Online platform posts and information * Key databases to link cryptographic keys * Public pictures or videos	*An open source review of the Facebook.com website for Gerren Johnson located a Facebook page for user "GREENJOHNSON88."* *Through additional social media research, pictures of PERSON were located on PERSON'S mother's Facebook page* *I was able to observe that it has been a vendor account since November 16, 2016 and has 531 confirmed sales of MDMA.*
Associated (non-government) accounts	Information from associated (non-government) accounts * Online platform accounts (i.e. cloud, email, social media) * Crypto exchange accounts * Prepaid cards * Physical accounts/registration (i.e. bank, post box, storage unit)	*The Twitter records also reflect the following different communications…* *According to subscriber information provided by Google* *Records obtained from Coinbase revealed that the account is subscribed in the name of*
Shared characteristics	Linking other information/ behavioral information * Fingerprint on letter * Linguistic expressions * Political views * Shared password across contexts * Similar package shipping style * Similar posting style * Timings fit to defendants schedule/ travel plans * Similar behavior	*I also reviewed the language in the terms and conditions and the refunds and dispute sections of the addy4cheap profile. The language was identical for addy4cheap's profile on both the Empire Market and Cryptonia* *H00k3d used the same avatar, a fish with a hook in it as depicted below, on at least the Dms Apollon, Avaris and Darkode*
Crypto tracing	Tracing cryptocurrency movements * Tracing crypto payments (i.e. Bitcoin)	*An analysis of Bitcoin transactions available via the public Blockchain reflects that the user of the MIMM Silk Road Account withdrew Bitcoins credited to the MIMM Silk Road Account to certain Bitcoin addresses (the "Ellingson Intermediary Bitstamp Bitcoin Addresses"). On several occasions in or about October 2013 and February 2014, Bitcoins were then transferred from the Ellingson Intermediary Bitstamp Bitcoin Addresses to the Ellingson Bitstamp Account.* *Based on a review of records available on the public blockchain, as well as a review of the Coinbase records, on January 6, 2020, approximately 16 minutes after receiving the deposit from the FBI undercover employee*
Undercover infiltration	Infiltrating with undercover agents or confidential informants * Agents crawling Onion Services * Agents buying products * Agents impersonating other entities (i.e. customers, money launderer) * Phishing and social engineering * Taking over accounts from criminals	*In or about mid-2014, an FBI online covert employee (the "OCE") assumed an online Dark-Web identity, which had previously been used by a trafficker in illicit materials, including, among other things, biological toxins* *Throughout the investigation, I, and other law enforcement agents, including the HSI-UC, have visited Silk Road 2.0 using undercover user accounts* *I have been involved in undercover purchases of narcotics from the website through an undercover account, which were ordered to and received in the Southern District of New York*

REFERENCES

Tippe, P., Tippe, A., 2024. Onion services in the wild: A study of deanonymization attacks. Proc. Priv. Enhancing Technol. 2024, 291–310. URL: https://doi.org/10.56553/popets-2024-0117, doi:10.56553/POPETS-2024-0117.